VIT

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

CHENNAI

22BAI1286

Final Assessment Test(FAT) - Apr/May 2025

Programme	**B.Tech.**	Semester	**Winter Semester 2024-25**
Course Code	**BCSE417L**	Faculty Name	**Prof. Rajarajeswari S**
Course Title	**Machine vision**	Slot	**A1+TA1**
		Class Nbr	**CH2024250502405**
Time	**3 hours**	Max. Marks	**100**

Instructions To Candidates

- Write only your registration number in the designated box on the question paper. Writing anything elsewhere on the question paper will be considered a violation.

Course Outcomes

CO1: Understand the basics of how an image is processed
CO2: Enhance, Analyze and segment the image using algorithms
CO3: To interpret the image and transform it using the mathematical knowledge
CO4: Extract the features from the image and represent using morphological operations
CO5: Apply the concept in understanding the scene and process the background part of the image

Answer all Questions (10 × 10 Marks)

01. **a.** A convex rearview mirror of a car has a focal length of -30cm and is used to detect nearby vehicles. A car is 2m away from the mirror. Determine the image distance, and if the driver wants a wider field of view, should the focal length of the convex mirror be increased or decreased? Justify with ray diagram analysis. **(5 Marks)**

b. Outline why Hue (H) from the HSV color model is more suitable than RGB values for vegetation classification under different lighting conditions. Convert an RGB pixel (R=100, G=200, B=50) into its HSV representation using a suitable conversion formula. **(5 Marks)**

[10] (CO1/K2)

02. A traffic surveillance system struggles with foggy conditions, reducing the clarity of vehicles and road signs. The original 5×5 grayscale matrix and the histogram for clear visibility are given below

Grayscale Matrix

$$\begin{bmatrix} 25 & 30 & 35 & 30 & 25 \\ 30 & 40 & 50 & 40 & 30 \\ 35 & 50 & 60 & 50 & 35 \\ 30 & 40 & 50 & 40 & 30 \\ 25 & 30 & 35 & 30 & 25 \end{bmatrix}$$

The histogram for clear visibility:

Target Intensity	Required Frequency
15 (Darkest Areas - Road Shadows)	5
35 (Vehicles - Mid-Tone Regions)	7
55 (Road Signs - High Contrast)	6
75 (Headlights/Brighter Areas)	7

a. Compute the mapping function to transform the foggy image to match the clear visibility target. **(8 Marks)**
b. Justify how the transformation improves vehicle and road sign detection. **(2 Marks)**

[10] (CO2/K3)

03. A drone surveillance system captures a low-resolution thermal image of a disaster-affected area to analyze the heat signatures of survivors. However, due to sensor limitations, some pixels are missing. The team must estimate the missing values by linearly interpolating in both directions to enhance image clarity before further processing. The given 4×4 matrix represents thermal intensity values (in °C) at different pixel locations:

	0	1	2	3
0	30	35	40	45
1	38	44	-	50
2	45	55	-	55
3	50	52	54	58

Estimate the missing pixel intensity at (1,2) and (2,2) and compute the intensity at (1.5,2.5) with the calculated value. Discuss how this method affects image accuracy.

[10] (CO2/K3)

04. A 4x4 grayscale image represents depth variations from a ground-penetrating radar (GPR) scan used to detect underground water reservoirs.

$$\begin{bmatrix} 10 & 14 & 18 & 22 \\ 7 & 12 & 16 & 21 \\ 5 & 9 & 13 & 18 \\ 2 & 6 & 10 & 15 \end{bmatrix}$$

a. Apply a two-stage split process:
- Split the given image into 2×2 regions where $\sigma > 4$.
- Further split sub-regions where $\sigma > 2.5$. **(4 Marks)**

b. Apply the merging technique for regions where mean differences are < 5. **(4 Marks)**

c.Identify and mark the subsurface water zones where the intensity values are significantly lower than surrounding regions. **(2 Marks)**

[10] (CO3/K3)

05. A binary image (7×7) contains small, isolated noise points (single pixels = 1) in the background and some small holes (0s) inside objects. The following morphological operations are applied in sequence using a 3×3 Plus-shaped structuring element:
- Operation X reduces small noise points but slightly shrinks the object.
- Operation Y restores the object's shape and fills small holes inside it.

Given Image

$$\begin{bmatrix} 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 & 0 \end{bmatrix}$$

a. Compute the resultant images after Operation X and Operation Y. **(8 Marks)**

b. Analyze how changing the size of the structuring element affects the result in morphological image processing **(2 Marks)**

[10] (CO4/K2)

06. Consider the following 5×5 grayscale image matrix, where the pixel value **65** appears at different positions:

	0	1	2	3	4
0	30	40	50	55	60
1	45	65	55	65	75
2	50	75	80	85	90
3	60	65	75	95	100
4	70	85	100	105	110

a. Compute Standard LBP for 65 at (1,1) and (1,3) using a 3×3 LBP operator (radius = 1) **(4 Marks)**
b. Compute weighted LBP values for both positions with weight (diagonal) = 2 and weight (horizontal/vertical) = 1 **(4 Marks)**
c. Do the two pixels with the same intensity (65) have identical LBP values? Justify with reasons. **(2 Marks)**

07. Given a 4x4 grayscale image matrix: **[10] (CO4/K4)**

4	6	4	4
6	4	10	6
4	10	4	6
4	6	10	6

a. Apply the Haar wavelet transform to compute the approximation and detail coefficients. **(4 Marks)**
b. Perform the inverse Haar transform to reconstruct the original image and verify the correctness. **(4 Marks)**
c. Discuss how Haar wavelets can be used for edge detection and why they are effective for certain image compression techniques. **(2 Marks)**

[10] (CO3/K3)

08. Consider a scenario where you have a set of multi-modal medical images that have been obtained using different medical imaging modalities like (e.g., MRI, CT, PET) of a patient's brain.
a. Describe the advantages and challenges associated with fusing these multi-modal images, especially in the healthcare domain. **(6 Marks)**
b. Design a robust image fusion algorithm that effectively combines the complementary information from these modalities to enhance the diagnostic capabilities and also write necessary equations and architecture. **(4 Marks)**

[10] (CO1/K2)

09. Suppose that you are developing a system for 3D reconstruction. Elucidate the role of depth information for the current scenario and a methodology to optimize depth estimation from epipolar geometry by intelligently combining geometric constraints with efficient deep learning architectures, enabling accurate and robust depth recovery even in the presence of challenging image conditions and dynamic scene variations.

[10] (CO5/K4)

10. Imagine that you are developing a machine learning algorithm that relies on extracting local image features that are invariant to significant variations in scale, rotation, illumination, and viewpoint. Propose a robust approach for constructing a local feature descriptor that captures fine-grained details while maintaining robustness to geometric and photometric transformations. Explain all the steps in detail with the necessary equations.

[10] (CO5/K3)

BL-Bloom's Taxonomy Levels - (K1-Remembering,K2-Understanding,K3-Applying,K4-Analysing,K5-Evaluating,K6-Creating)

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